Batterjee Medical College
For Science and Technology

Adrenal Gland Disorders

Internal Medicine Department

Knowledge

- Relate pathophysiological basics and aetiology of adrenal gland disorders
- Describe epidemiology, manifestations, complications and management of hyper and hypoadrenalism
- Describe causes, manifestations of hirsutism

Skills

- Interpret clinical manifestations and investigations of adrenal gland disorders to formulate reasonable diagnosis.
- Apply evidence-based management plan for adrenal gland disorders
- Distinguish patients in emergency situations to formulate a comprehensive management plan; ensuring keeping patients in normal homeostasis

Main Subtopics

- Adrenal insufficiency/Addison's disease (causes, clinical picture, investigations, differential diagnosis)
- Adrenal crisis (clinical picture and emergency management)
- Congenital adrenal hyperplasia (17α-, 21α-, and 11β-hydroxylase deficiencies: hormonal changes and clinical features)
- Phaeochromocytoma (catecholamine synthesis, clinical picture, complications, investigations, treatment)
- Male hypogonadism (diagnostic approach and treatment)
- Hirsutism (basic hair physiology, diagnostic approach, and treatment)

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ADDISON'S DISEASE
CHRONIC ADRENAL FAILURE

Causes

A. Primary:

1.Auto-immune adrenalitis : (most common cause, 80 % of 1ry)

This may occur alone or in combination with other auto-immune endocrine insufficiency including **IDDM**, **hypothyroidism** (& sometimes hypogonadism), hypoparathyroidism or pernicious anemia "Schmidt syndrome".

2.TB of the adrenal glands (10 - 20 %).

3.Following bilateral adrenalectomy & congenital enzymatic defects.

4.Sarcoidosis, secondaries, amyloidosis or hemochromatosis.

B. .Secondary to hypopituitarism (Simmonds disease).

C. Tertiary adrenal insufficiency: caused by conditions that decrease CRH production. The most common cause is sudden discontinuation of chronic glucocorticoid therapy.

Clinical pearl

Secondary and tertiary adrenal insufficiency are far more common than primary adrenal insufficiency!

Factors Affecting the Adrenals

LIFE ISSUES:

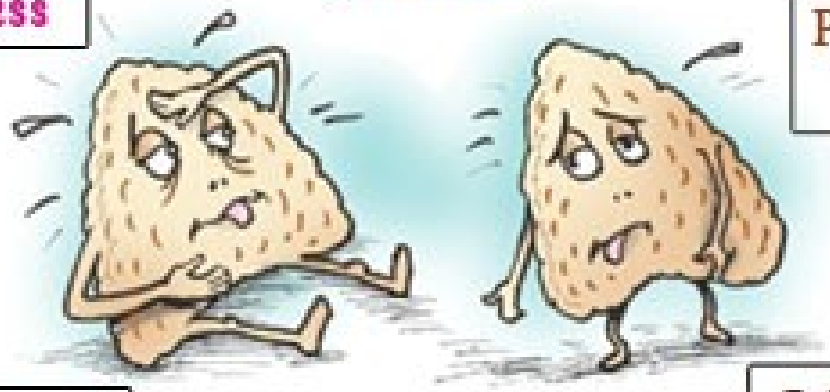
- Financial Pressures
- Death of a Loved One
- Emotional Stress
- Unwanted Unemployment
- Marital Stress
- Psychological Stress

Negative Attitudes & Beliefs

FEAR

BODY ISSUES:

- Allergies
- Infections: Acute & Chronic
- Prescription Drugs
- Wound Healing



LIFESTYLE

- Smoking
- Lack of Sleep
- Lack of or Excessive Exersise
- Over Exertion
- Lack of RELAXATION

TOXINS

DIET

- COFFEE
- Lack of Good Food
- CAFFEINE
- WHITE FLOUR PRODUCTS
- SUGAR

POOR EATING HABITS

Clinical pictures

- Onset is usually insidious mostly in the adult life & the course. is slowly progressive.

1.Asthenia:

- Weakness of muscles. (+ muscles cramps due to hyponatraemia)
- Weight loss.

o Due to

- Lack of cortisol.
- Hyperkalemia.
- Hypoglycemia.

2.Hypotension

- Systolic blood pressure → 110 mmHg excludes Addison's disease.
- Postural hypotension → common due to hypovolaemia & dehydration.

3.Hypoglycaemia: leading to :

Drowsiness, hunger pain & even coma.

4.Hyperpigmentation of skin & mm.:



- ***Sites :***
  - Face, neck, groins, axillae, nipples, friction areas, scars as umbilicus, areas exposed to sunlight.
  - mm. of mouth, tongue; rectum







- ***Causes :***
  - Hypocorticism → ↑ ACTH → ↑ melanocyte stimulating hormone (& lipotropin) → ↑ pigmentation.
  - Destruction of the adrenal medulla: → diversion of tyrosine to form melanin instead of adrenaline & noradrenaline.

NB: - Hyperpigmentation isn't a feature of 'Simmonds' disease.

5.- Loss of adrenal androgen:

- ↓ Axillary & pubic hair in ♀ (but in ♂, testicular androgen is preserved).
- Infertility & amenorrhea in females.

6.GIT disturbances:

- Nausia, vomiting, diarrhea (& even steatorrhea).

7.Polyuria: due to Na diuresis.

There may be tenderness in the renal angle in TB of adrenals.

N.B: the patients may present with Addisonian crisis.

Clinical pearl

Signs of aldosterone deficiency can help differentiate primary adrenal insufficiency from secondary and tertiary adrenal insufficiency.

Hypocortisolism

Fatigue, lethargy

Orthostatic hypotension

Weight loss

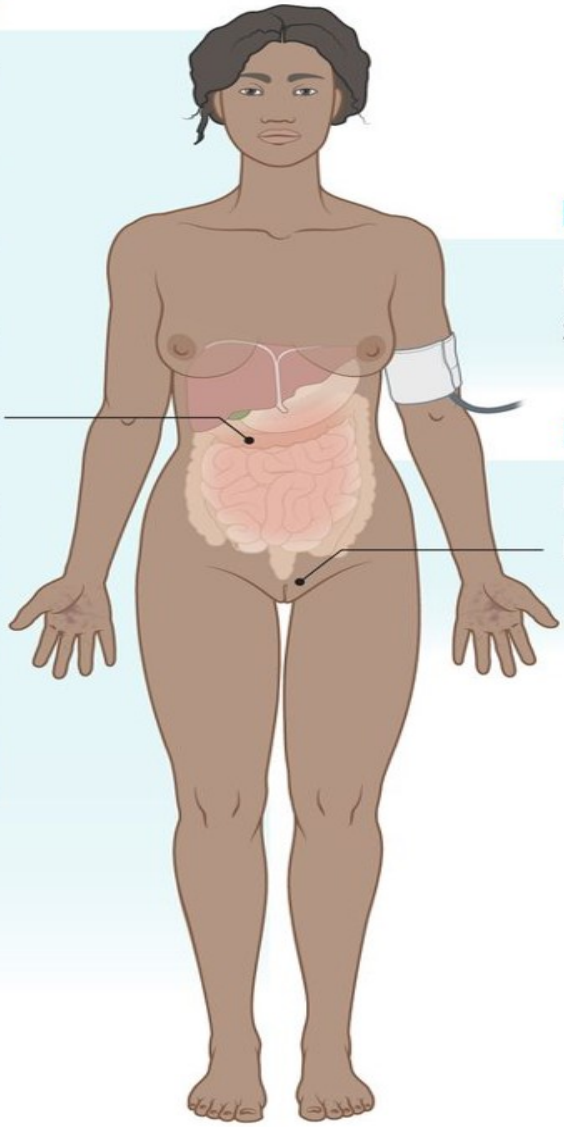
Sugar craving

GI disturbances

Hyperpigmentation, including unexposed areas (e.g., palmar creases, intraoral mucosa)







Hypoaldosteronism

Hypotension

Salt craving

Hypoandrogenism

Loss of libido

Loss of axillary and pubic hair

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Hormonal changes	Clinical features	Laboratory findings	Primary adrenal insufficiency	Secondary adrenal insufficiency	Tertiary adrenal insufficiency
Hypoaldosteronism	• Hypotension • Salt craving	• Hyponatremia • Hyperkalemia • Normal anion gap metabolic acidosis	✓	• Absent	• Absent
Hypocortisolism	• Weight loss, anorexia • Fatigue, lethargy, depression • Muscle aches • Weakness • Gastrointestinal complaints (e.g., nausea, vomiting, diarrhea) • Sugar cravings • (Orthostatic) hypotension	• Hypoglycemia • Hyponatremia	✓	✓	✓
Hypoandrogenism	• Loss of libido • Loss of axillary and pubic hair	• ↓ DHEA-S	✓	✓	✓

Investigations

1. ↓ plasma cortisol level.
2. Morning ACTH level: Obtain if morning cortisol is low.
   - Primary adrenal insufficiency: elevated ACTH levels
   - Secondary/tertiary adrenal insufficiency: ACTH levels low to normal
3. ↓ 24 hour urinary cortisol, l7 oHCs or 17 Ks (ketosteroids).
4. ACTH stimulation test " (cosyntropin test): gold standard to confirm adrenal insufficiency) (I. M or I. V.)
   - Normal response → plasma cortisol ↑ by 6 ug / dl above the basal value
   - In 1ry adrenal insufficiency → no response
   - In 2ry adrenal insufficiency →normal response
   - In 3ry, no or mid response (atrophy of pit. cells from no stimulation).
5. Biochemical changes :
   - ↓ serum Na, Cl.
   - ↑ serum K, acidosis.
   - ↑ serum Ca.
   - ↓ serum glucose.

6.Urine:

- ↑ volume.
- ↑ Na & Cl & ↓ K excretion.

7.Blood picture :

- Normochromic anaemia.
- Haemoconcentration.
- Eosinophilia & lymphocytosis.

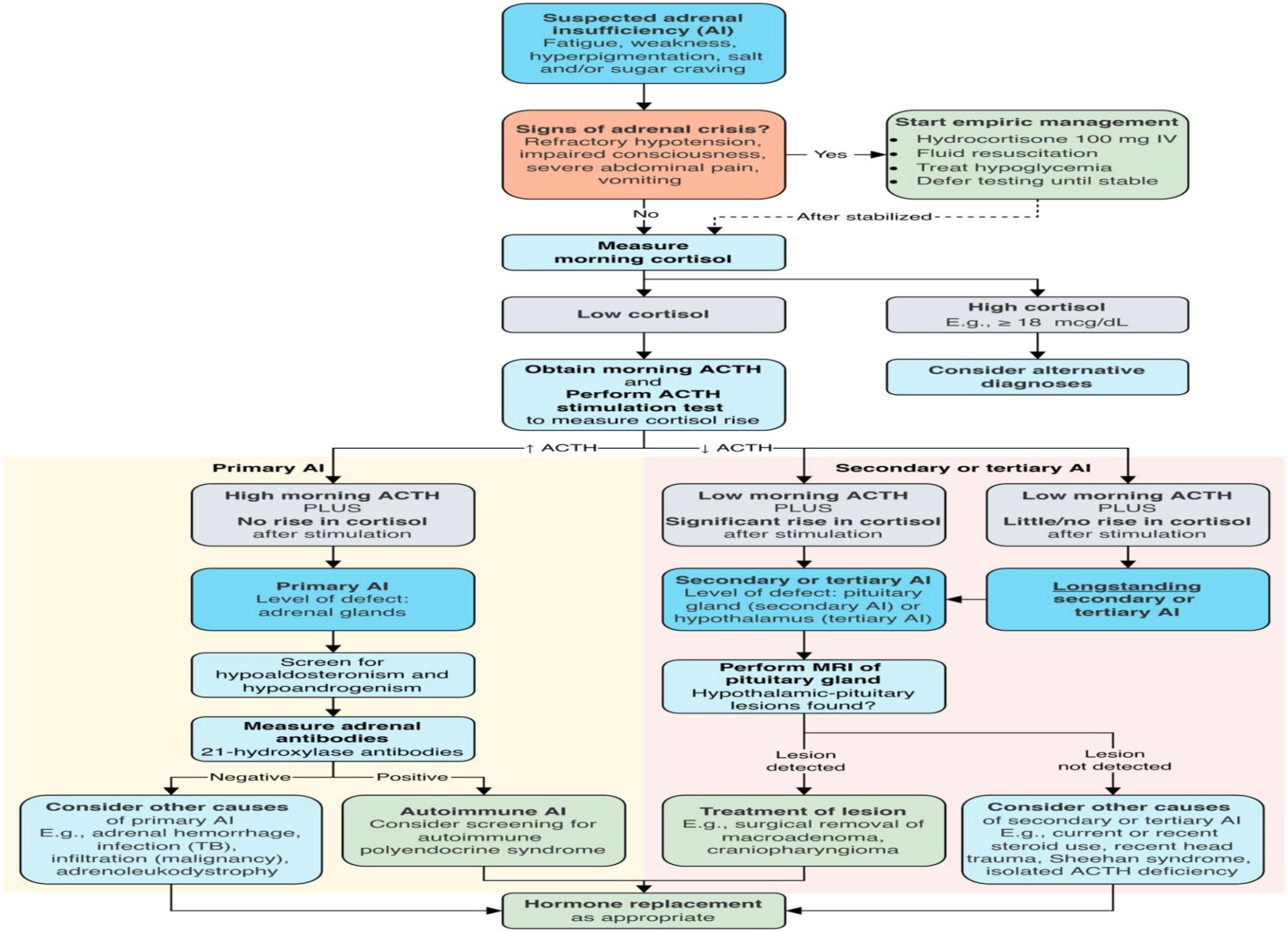
8.Investigation to find the cause :

- Adrenal antibodies in → autoimmune Addison.
- X-ray: - adrenal calcification in T.B. cases.
  - Evidence of pulmonary TB.

	Morning cortisol	Morning ACTH	ACTH stimulation test
Primary adrenal insufficiency	↓	↑	No increase in serum cortisol after stimulation
Secondary/tertiary adrenal insufficiency		↓	Increase in serum cortisol after stimulation
Long-standing secondary/ tertiary adrenal insufficiency [18]		↓	No (or very little) increase in serum cortisol after stimulation

⚠ Caution

Maintain a low threshold for adrenal insufficiency screening (especially in patients with nonspecific symptoms), as up to 50% of patients present with adrenal crisis, often due to missed or delayed diagnosis. [19]


Suspected adrenal insufficiency (AI)
Fatigue, weakness, hyperpigmentation, salt and/or sugar craving
Signs of adrenal crisis?
Refractory hypotension, impaired consciousness, severe abdominal pain, vomiting
Yes
Start empiric management
Hydrocortisone 100 mg IV
Fluid resuscitation
Treat hypoglycemia
Defer testing until stable
No
After stabilized
Measure morning cortisol
Low cortisol
High cortisol
E.g., ≥ 18 mcg/dL
Obtain morning ACTH and Perform ACTH stimulation test to measure cortisol rise
Consider alternative diagnoses
↑ ACTH
↓ ACTH
Primary AI
Secondary or tertiary AI
High morning ACTH PLUS No rise in cortisol after stimulation
Low morning ACTH PLUS Significant rise in cortisol after stimulation
Low morning ACTH PLUS Little/no rise in cortisol after stimulation
Primary AI
Level of defect: adrenal glands
Secondary or tertiary AI
Level of defect: pituitary gland (secondary AI) or hypothalamus (tertiary AI)
Longstanding secondary or tertiary AI
Screen for hypoaldosteronism and hypoandrogenism
Perform MRI of pituitary gland
Hypothalamic-pituitary lesions found?
Measure adrenal antibodies
21-hydroxylase antibodies
Lesion detected
Lesion not detected
Negative
Positive
Consider other causes
of primary AI
E.g., adrenal hemorrhage, infection (TB), infiltration (malignancy), adrenoleukodystrophy
Autoimmune AI
Consider screening for autoimmune polyendocrine syndrome
Treatment of lesion
E.g., surgical removal of macroadenoma, craniopharyngioma
Consider other causes
of secondary or tertiary AI
E.g., current or recent steroid use, recent head trauma, Sheehan syndrome, isolated ACTH deficiency
Hormone replacement
as appropriate

Differential diagnosis

1.DD. from 2ry hypoadrenalism .

2.Other causes of pigmentation cg. racial, hemochromatosis, porphyria & ancanthosis nigricans, neurofibromatosis.

3.Asthenia & pigmentation may occur also in: e.g. Pellagra, CML (especially, in patients on busulfan therapy), CRF, chronic arsenic poisoning, thyrotoxicosis, advanced malignancy, malabsorption syndrome.

Treatment

A. Replacement therapy :

- Glucocorticoids →prednisolone 5 mg in the morning & 2.5 mg in the evening.
- Mineralocorticoids → fludrocortisone 0.1 mg every other day.

N.B: Active TB. cases should receive anti-TB therapy.

B. Management during stressful conditions :

Minor illness	Major illness	
e.g. - Dental extraction. - Upper respiratory tract infection • The dose is doubled "until the condition is resolved".	For elective major surgery	For catastrophies & emergencies
	• 100 mg hydrocortisone IV during the night preceedmg surgery & then 100 mg / 8 hours. • Then →the dose is gradually tapered within 3 — 5 days to reach the previous dose.	e.g. - trauma, - myocardial infarction - major emergency surgery, …etc → ttt as acute adrenal failure.

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ACUTE ADRENAL FAILURE
ADDISONIAN CRISIS

Causes

1. Previously undiagnosed patients may present in crisis either spontaneously or precepitated by intercurrent illness.

2. Patients already on cortisone, on facing severe intercurrent illness without increasing the replacement dose.

3. Sheehan syndrome: if ttt is initiated with thyroxin alone without cortisone.

4.After bilateral adrenalectomy (or pituitary surgery).

5.After sudden corticosteroid withdrwal.

6.Acute meningococcal septicaemia (waterhouse friderichsen syndrome). Hge may occur in the adrenal gland → acute adrenal failure.

7.Medical ttt for adrenal carcinoma (eg. with metyrapone, mitotane).

8.Massive thrombosis of adrenal veins (pregnancy, puerperium & burns).

Diagnosis

Clinical picture:

a. Sudden onset of severe weakness, mental confusion, abdominal pain, anorexia, nausea, vomiting, diarrhea.

b. Dehydration → hypotension, shock.

c. If not treated → coma & death within 24 hours.

> ⚠ Caution
>
> Most cases of adrenal insufficiency are subclinical and only become apparent during periods of stress (e.g., surgery, trauma, infections), when the cortisol requirement is higher!

Consider adrenal crisis in patients with severe hypotension refractory to fluid resuscitation and/or vasopressors.

Investigations:

- ↓ Serum Na.
- ↑ Serum K.
- ↓ Serum cortisol.

Treatment

1.Correction of volume depletion & electrolyte status by I.V. saline & glucose "10 %".

2.Hydrocortisone (Solucortif) + 100 mg I.V. then 100 mg as I.V. infusion / 8 hours(in 5 % glucose) → then gradual tapering (until the maintenance dose is reached within 5 days).

3.Fludrocortisone: →0. l mg / day (added when the daily dose of hydrocortisone has reached 100 mg / day).

4.Proper ttt of infections & precipitating factor.



The **5 S**'s of adrenal crisis treatment are **Salt** (0.9% saline), **Sugar** (50% dextrose), **Steroids** (100 mg hydrocortisone IV once, then 200 mg over 24 hours), **Support** (normal saline to correct hypotension and electrolyte abnormalities), and **Search** (for the underlying disorder).

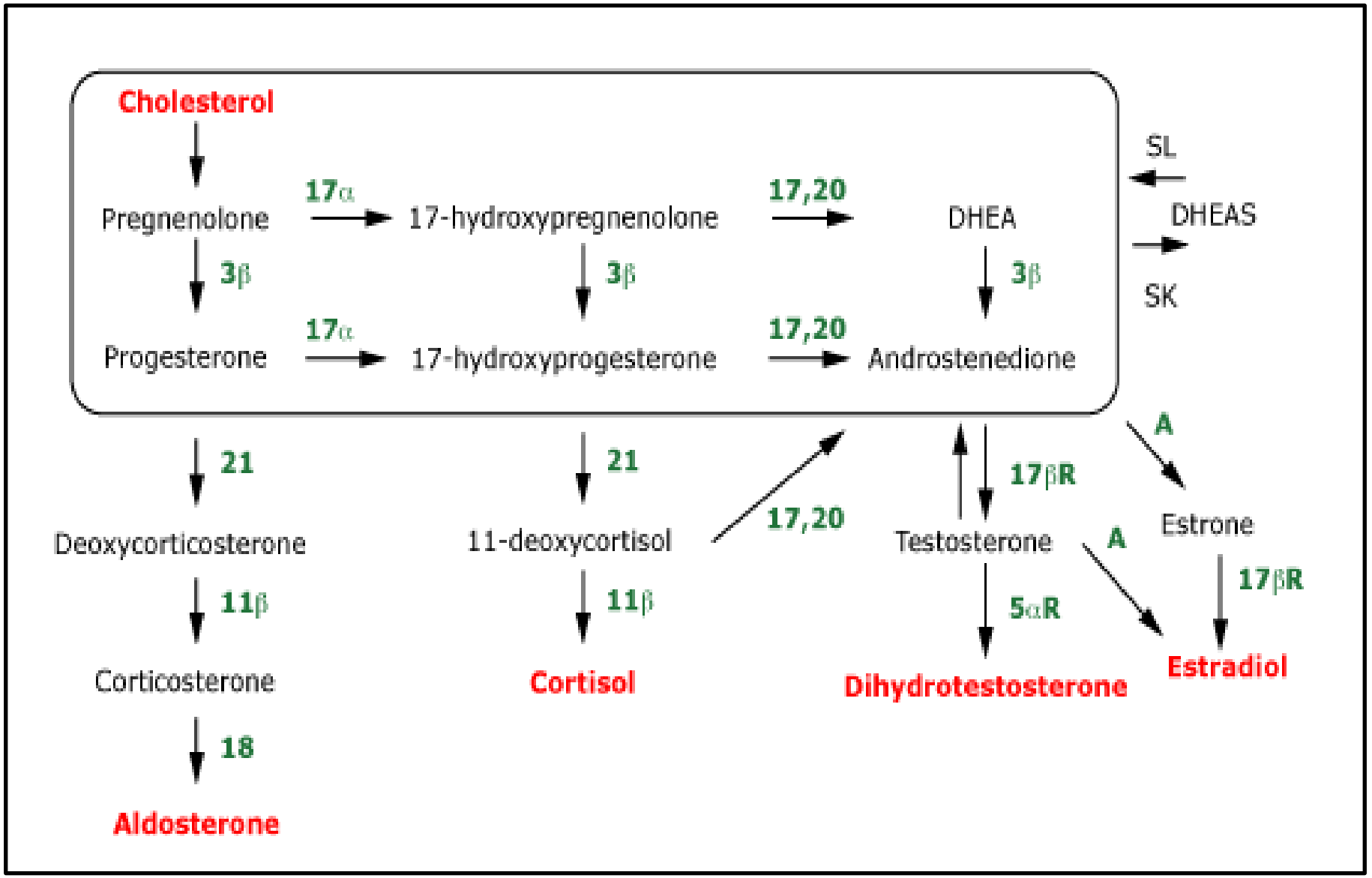
⚠ Caution

Adrenal crisis can be life-threatening, so treatment with high doses of hydrocortisone should be started immediately, without waiting for diagnostic confirmation of hypocortisolism!

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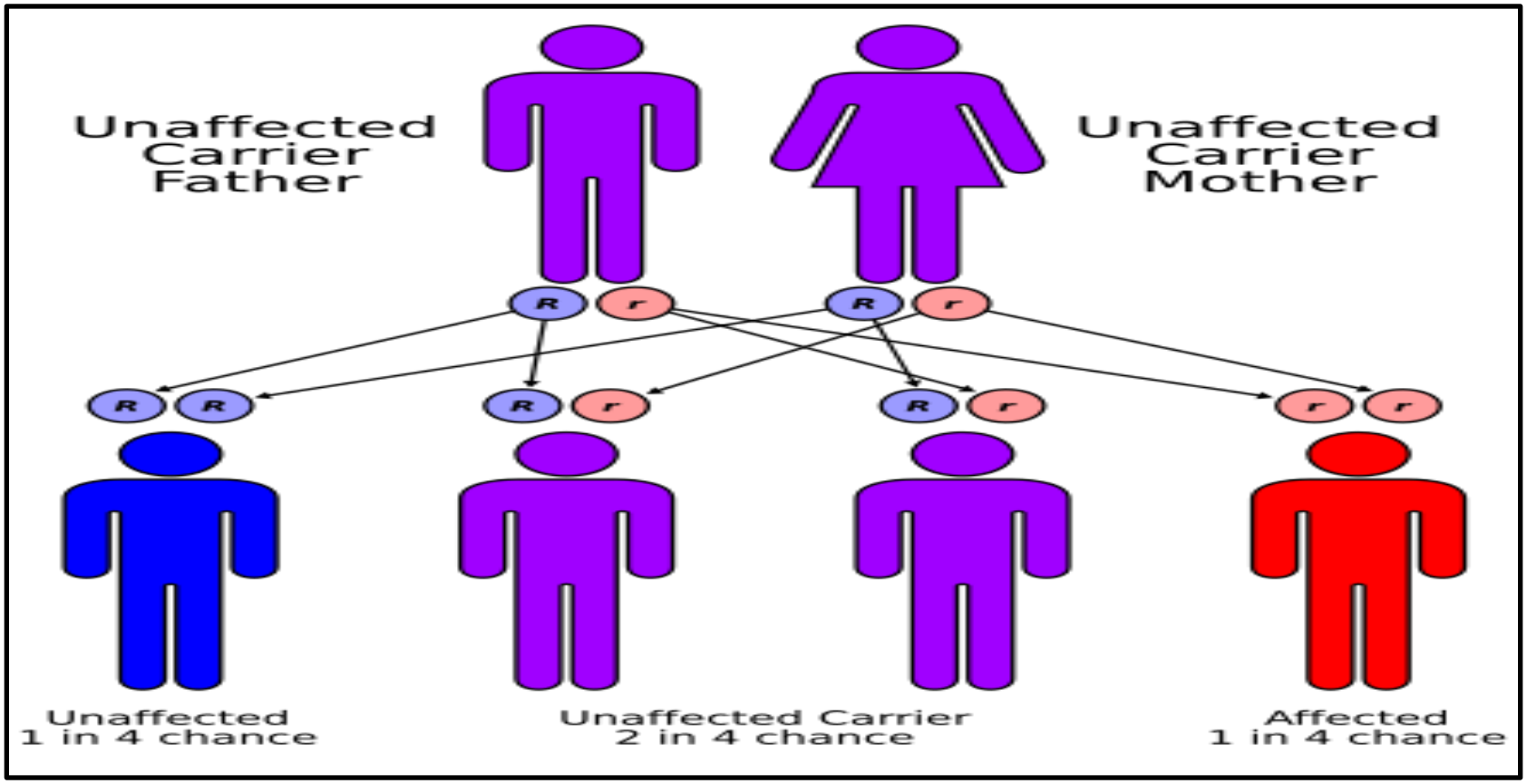
CONGENITAL ADRENAL HYPERPLASIA
" ADRENOGENITAL SYNDROME "

Synthesis of Adrenal Hormones



Definition

- Group of conditions caused by inherited enzymatic defects in the enzymes necessary for the synthesis of cortisol → ↓ cortisol → ↑ ACTH adrenal hyperplasia & ↑ synthesis of the hormone proximal to the block.



Causes/CP

Enzyme ↓	Hormonal changes	Clinical features
17 oH ase ↓ extremely rare	-↓ glucocorticoids & sex hormones. - ↑ mineralocorticoids	• **HPN & Hypokalemia** • In ♀→ **sexual infantilism** • In ♂→ ♂ **pseudohermaphroditism**
21 oH ase ↓ it is the commonest form	- ↓ glucorticoids & mineralocorticoids. - ↑ sex steroids	• Salt losing syndrome: due to aldosterone ↓: o **Hypotension** o **Hyperkalemia** • Androgen excess: o ♀ **pseudohermaphroditism**: (big clitoris, small vagina, the libia majora are big & fused simulating scrotum+normal internal jenitalia) o ♂ **precocious puberty:** (Macrojenitosmia precox), but without spermatogenesis & the testicle remain small, masculinization, deeping of voice & premature fusion of epiphysis dwarfism
11 B oH ase ↓ about 10% of cases	- ↑ 11 deoxycorticosterone, deoxycortsol & androgns. - ↓ cortisol	• **Hypertension** • Virilization → ♀ **pseudohermaphroditism** • **Precocious puberty**

Treatment

1.Replacement therapy:

- Prednisolone (7.5 mg/ day).
  - To replace the deficient cortisol in glucocorticoids deficient forms.
  - ↓ ACTH :
    - ↓ Blood pressure in hypertensive forms.
    - ↓ Androgen in virilizing forms.
- Fludrocortisone (0. 1 mg / day) in mineralocorticoid deficient forms.

2.Surgical correction :

- For ♀ pseudohermaphroditism .

Clinical pearl

Individuals with classic 21-hydroxylase deficiency require lifelong glucocorticoid replacement therapy.

Caution

Provide steroid stress dosing during severe infection, critical illness, and perioperatively to prevent adrenal crisis. [1]

N.B:

- Pseudohermaphroditism: → (phenotypic sex disturbance).
- True hermaphroditism:→ (chromosomal sex disturbance).

➢ (Differentiated by doing sex chromatin pattern)

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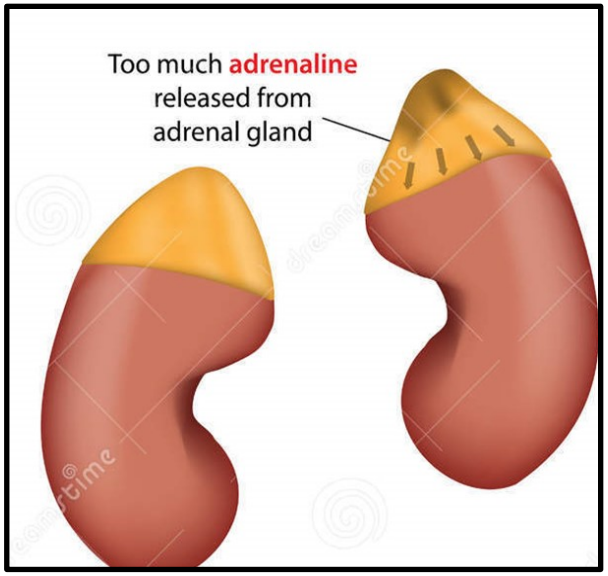
ADRENAL MEDULLA
PHEOCHROMOCYTOMA

It is a catecholamine — releasing tumour.

Commonly begin solitary adenoma (more on the right side).

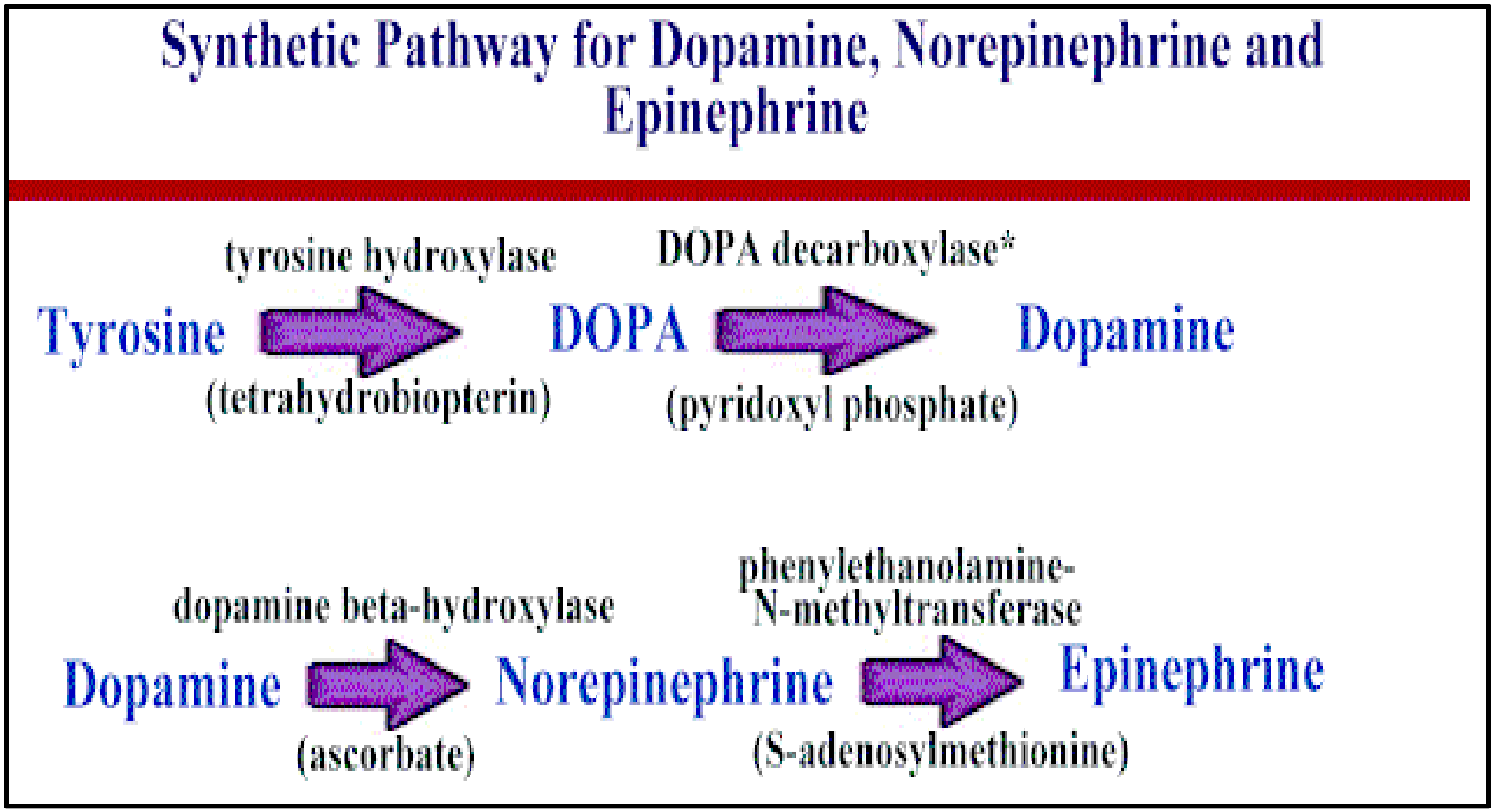
Rule of 10:

- 10 % are bilateral.
- 10 % are malignant.
- 10 % extra - adrenal.
- 10 % occur in children.



Familial Pheochromocytoma

- Occurs as a part of MEN II (Pheochromocytoma, thyroid medullary carcinoma & hyperparathyroidism" Sipple's syndrome").



- Norepinephrine is secreted by most of adrenal tumours & all extra-adrenal tumours.
- Epinephrine → secreted by adrenal tumours mainly

Clinical manifestations

- Headaches, rapid heartbeat, excessive sweating, and anxiety. These symptoms can occur in episodes that may last for several minutes or longer, and can vary in severity and frequency.
- Other potential symptoms include nausea, vomiting, tremors,, and chest or abdominal pain.
- Cold hands & feet pale skin& pallor of the face. →due to Vasoconstriction (α -stimulation)
- High blood pressure (Hypertension),paroxysmal → in 50 % of cases, while in the other 50 % it is persistent. Postural hypotension may occur.

Clinical pearl

Patients can show typical clinical features (e.g., secondary hypertension) but may also be asymptomatic.

Student pearl

5 most important Problems (5 P's) of Pheochromocytoma: increased blood Pressure, head Pain (headache), Perspiration, Palpitations, and Pallor

- Complications of hypertension on C.V.S. & C.N.S. (……..)
- Constipation & ischaemic enterocolitis (GIT manifestations).
- Masses: in the abdomen, neck.
- Metabolic : diabetogenic due to :
  - β - effect: ↑ glycogenolysis.
  - α - effect:↓ insulin secretion
- Factors precipitating the attacks:
  - Pressure on the tumour.
  - Exercise.
  - Food e.g. cheese, alcohol.
  - Drugs e.g. beta—blockers.
  - Labour.

Investigations

1.Urine examination for:

a) Metanephrines (in 24 hours). (highly sensitive)
b) Free urinary catecholamines (in 24 hours).
c) Vanillyl mandelic acid (in 24 hours) : (N : 2.- 6 mg/day)

→False positive VMA may occur due to ingestion of vanilla, chocolate, aspirin.

2. Plasma free metaniphrines (highly sensitive)

3.Plasma catecholamine level.

4.Stimulation tests : Used in patients with infrequent episodes:

a) Glucagon : 1 mg IV. will provoke an attack in 90 % of cases.
b) Histamine : 50 ug I.V.

N.B: Phentolamine should be available during these 2 tests.

5.Suppression tests: Clonidine suppression test : (0.3 mg clonidine)

Plasma NA decreases to normal range 3 hours after receiving clonidine in essential HPN, not in Pheochromocytoma.



6.Localization of the tumour :

- o U/S.
- o CT
- o MRI
- o Radio-isotope scanning.
- o Sampling from SVC, IVC :

Determination of plasma catecholamines is useful in extra adrenal tumours that could not be determined by radiological methods.

Treatment

A. Surgical :

Na nitroprusside or phentolamine to stabilize blood pressure → then surgical removal of the tumour.

B. Medical :

1.Acute therapy : for patients with severe HPN

- Bed rest
- Phentolamine (regitine) → 2 - 5 mg IV. / 5 min.
- Na nitroprusside → 100 mg in 500 ml glucose 5%.
- Propranolol → 1 - 2 mg I.V / 5 minutes (to control arrhythmia) after adequate α adrenergic blockade.

Treat hypertensive crisis due to pheochromocytoma with IV antihypertensives that target catecholamine excess (e.g., phentolamine, clevidipine, or nicardipine) and avoid beta blockers. [17][18][19]

2.Prolonged medical therapy :

Indicated after acute therapy, if surgery to be delayed e.g. due to recent MI.

- phenoxybenzamine (Dibenzyline) :
  - Long acting α adrenergic blocker.
  - 10 mg twice/day (may be ↑ by 10-20 mg every day up to 40-200 mg / day).
- Propranolol : (after α blockade) → 20 - 40 mg / day.
- Metyrosine :

o Inhibitor of tyrosine oH ase enzyme.

o Can be used : Pre-operatively or In non operable cases.

Student pearl

Treat **PHE**ochromocytoma with **PHE**noxybenzamine, but remember that **A**lpha blockers should be given **A**head and **B**eta blockers **B**ehind.

Caution

Starting beta blockers before alpha blockers is contraindicated because doing so can cause life-threatening hypertensive crisis. Beta-blockers cancel out the vasodilatory effect of peripheral beta-2 adrenoceptors, potentially leading to **unopposed alpha-adrenoceptor stimulation** and thereby causing vasoconstriction and increased blood pressure.

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IMPORTANT NOTES ON REPRODUCTIVE SYSTEM

- Leydig cells →testosterone →↓LHRH → ↓ LH.
- Sertoli cells → Inhibin → ↓ FSH.
  →Mullerian inhibiting factor.

- The majority of circulating testosterone is bound to globulin (sex hormone binding globulin “SHBG ”).

→ **In the diagnosis of hypogonadism, notice :**

- ↓LH, FSH → pituitary causes (hypogonadotrophic hpogonadism).
- ↑ LH, FSH, ↓ testosterone → primary testicular disease (eg. irradiation, myotonia atrophica, cystic fibrosis, Klinefelter syndrome).
- Normal or ↑ LH, FSH, testosterone → end organ resistance.
- ↓ testosterone with no features of hypogonadism → ↓ SHBG.

→**In the treatment of hypogonadism:**

- Androgen replacement → IM, not orally → (liver toxicity).
- Pituitary causes → see before.
- Hypothalamic causes → LHRH (in pulsatile form ‘portable pump).

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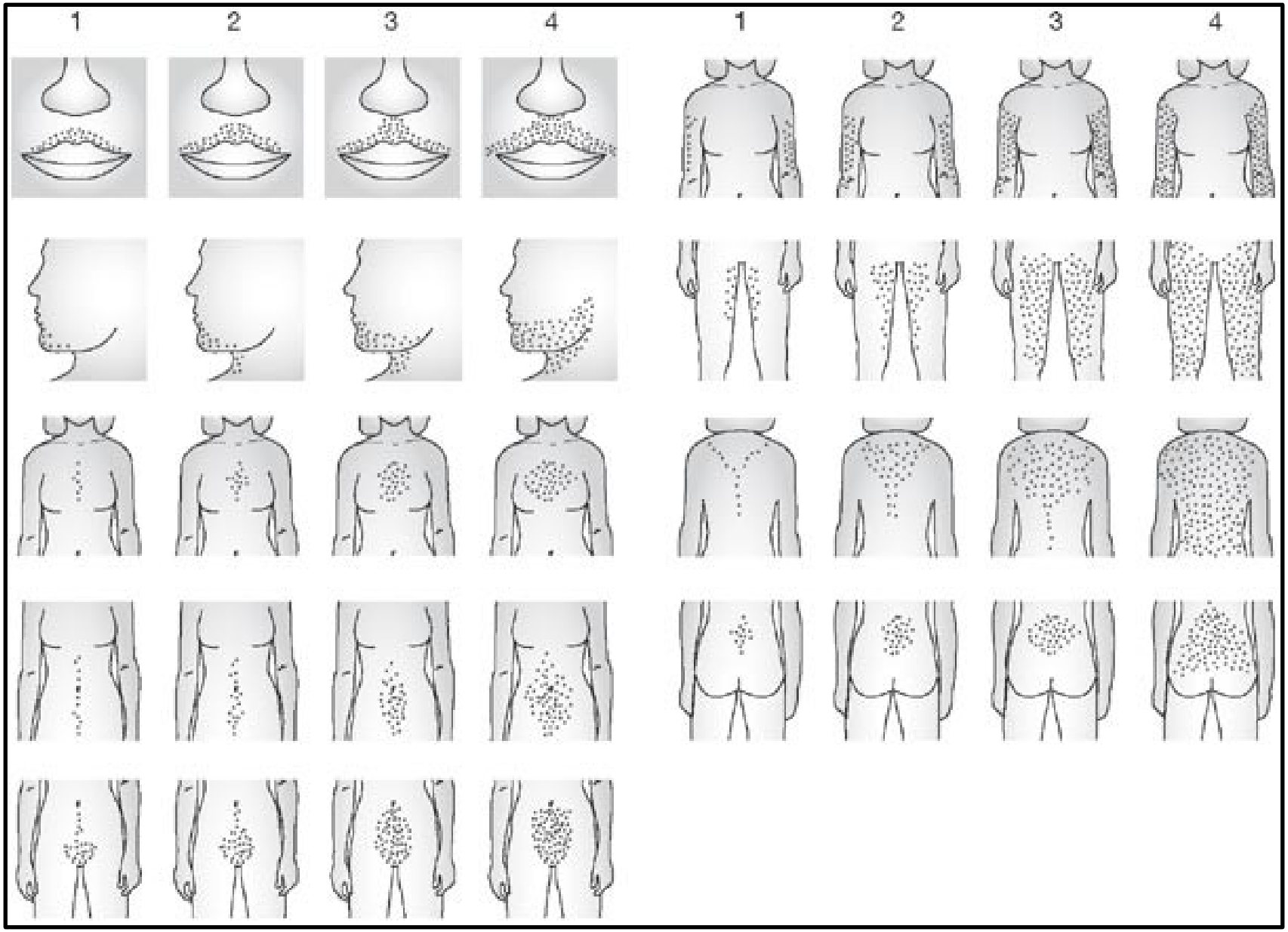
HIRSUTISM

Definitions:

- **Hirsutism** is a term reserved for females and refers to ↑ number of hair in areas associated with ♂ pattern hair growth e.g. face, chest, lower back, lower abdomen, inner thigh.
- **Hypertrichosis** is defined as excessive hair growth anywhere on the normal body hair distribution areas in either males or females.






1
2
3
4
1
2
3
4

Basic facts about hair

- Number of hair follicles is set from birth
- Hair grows from a individual hair follicle that are part of a pilosebaceous gland unit
- Main difference between sexes is the degree of differentiation of the hair
- Human hair growth is continuous
- Hair grows in a mosaic pattern (in a given area ,hair are in different stages of development)
- Some condition may cause a high level of synchrony between the growth cycles of hair ,leading to the appearance of either massive hair loss (alopecia)or excess hair for a limited period of time

Three types of Hair :

- Lanugo : Body hair seen in the fetus and newborn
- Vellus : Fine adult hair covering the body
- Terminal hair : Thick pigmented hair of scalp and pubic area

Thickness of the terminal hair varies form one individual to other depending upon genetic, and possibly nutritional

- Androgen sensitive hair : depend upon androgen input for hair growth → Face, neck, chest, abdomen, axillary, upper arms ,inner thighs and pubic hair, + part of the scalp hair.

- Less Androgen independent :

  Forearms ,hands .lower legs

Causes

1.Androgen—independent:

a. Constitutional, familial, racial.

b. Starvation. anorexia nervosa.

c. Drugs :

- o Minoxidil.
- o Phenytoin.
- o Diazoxide.

2.Androgen-dependent :

a. ovarian :

- Neoplastic: eg. arrenoblastoma.
- Non-neoplastic:

- Idiopathic hirsutism.
- Polycystic ovarian disease.

b. Adrenal:

- Neoplastic: virilizing adrenal tumour.
- Non-neoplastic: - Congenital adrenal hyperplasia.
- Cushing's syndrome.

c. Drugs: androgens

Treatment

1. Treatment of the cause: e.g.

- Stop the offending drug
- Treat tumours by surgical removal
- Treat medical disorders
- Weight reduction: 1st line treatment of PCOS in overweight women

2. Management directed to the target organ and cells

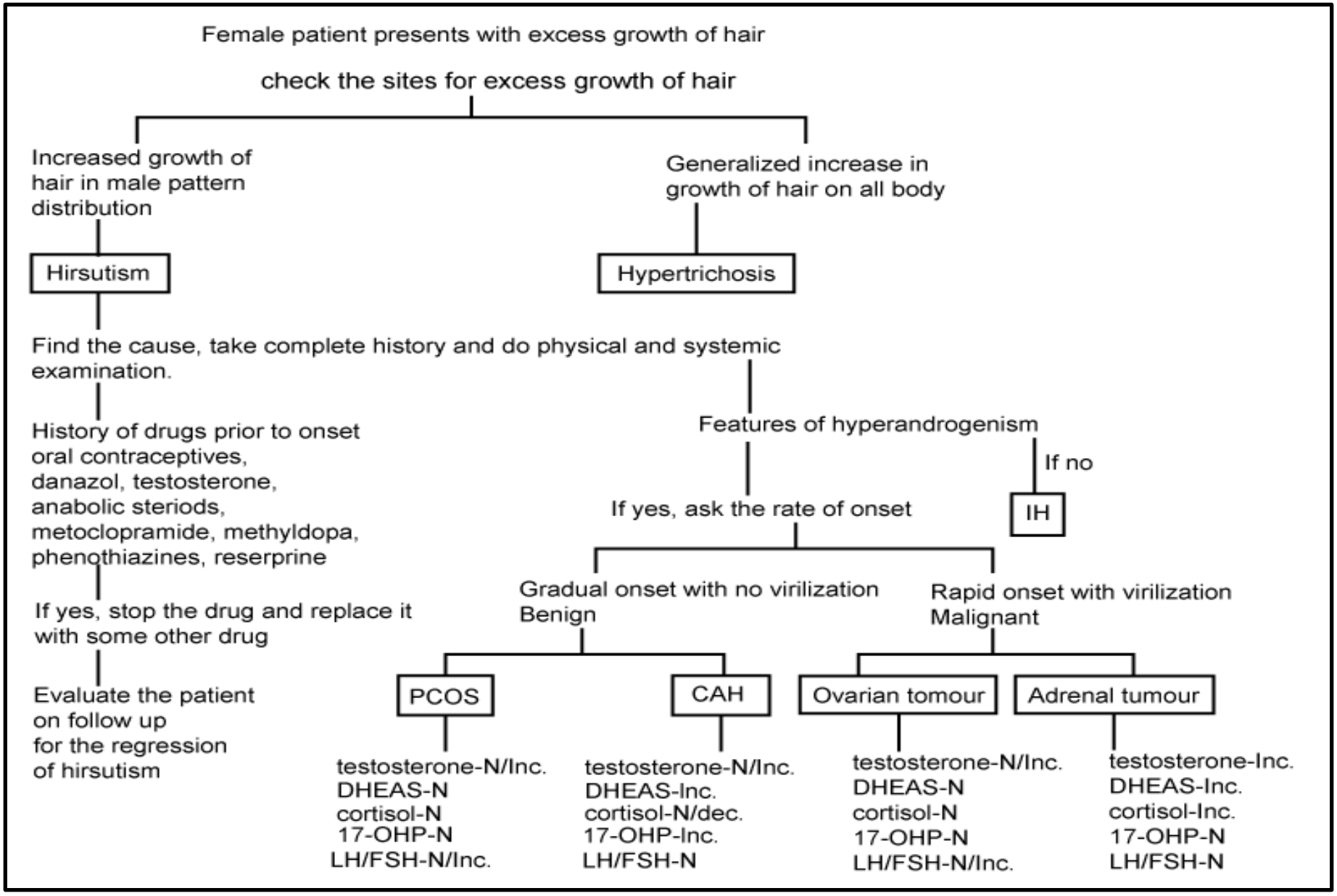
- Competition with Androgen receptors at hair follicles
- Spironolactone, Flutamide, Ketoconazole, Cyproterone acetate
- 5-alpha reductase Inhibitors : Finasteride

3. Recent : Eflornithien: vaniqua (13.9% cream)

- It irreversibly inhibits ornithine decarboxylase (ODC), an enzyme that catalyzes the rate-limiting step for follicular polyamine synthesis, which is necessary for hair growth.
- Improvement occurs gradually over a period of 4-8 weeks or longer.
- Most reported adverse reactions consisted of minor skin irritation.

4. Cosmotic treatment: by

- Bleaching :- can cause irritation, purities, skin discoloration
- Shaving
  - Shaving does not lead to worsening of hirsutism and is a good short-term solution for facial hair.
  - Does not affect the rate or duration of anagen phase, or diameter of hair but yields a blunt tip – illusion of thicker hair
- Plucking, Waxing - : scarring, folliculitis, hyperpigmentation
- Depilatory creams - Irritant dermatitis
- Electrolysis - painful, erythema, inflammation, scarring
- Laser


Female patient presents with excess growth of hair
check the sites for excess growth of hair
Increased growth of hair in male pattern distribution
Generalized increase in growth of hair on all body
Hirsutism
Hypertrichosis
Find the cause, take complete history and do physical and systemic examination.
History of drugs prior to onset oral contraceptives, danazol, testosterone, anabolic steriods, metoclopramide, methyldopa, phenothiazines, reserprine
If yes, stop the drug and replace it with some other drug
Evaluate the patient on follow up for the regression of hirsutism
Features of hyperandrogenism
If no
IH
If yes, ask the rate of onset
Gradual onset with no virilization
Benign
Rapid onset with virilization
Malignant
PCOS
CAH
Ovarian tomour
Adrenal tumour
testosterone-N/Inc.
DHEAS-N
cortisol-N
17-OHP-N
LH/FSH-N/Inc.
testosterone-N/Inc.
DHEAS-Inc.
cortisol-N/dec.
17-OHP-Inc.
LH/FSH-N
testosterone-N/Inc.
DHEAS-N
cortisol-N
17-OHP-N
LH/FSH-N/Inc.
testosterone-Inc.
DHEAS-Inc.
cortisol-Inc.
17-OHP-N
LH/FSH-N

Case Scenario

A 30-year-old woman comes to the physician because of a swelling on her neck for 5 months. It has gradually enlarged in size and is mildly painful. She has also had intermittent episodes of throbbing headache, sweating, and palpitations over the past 3 months. Menses occur at regular 28-day intervals and last for 4–5 days. She does not smoke and occasionally drinks alcohol on weekends. She appears thin and pale. Her temperature is 38.7°C (101.7°F), pulse is 112/min, and blood pressure is 140/90 mm Hg. Examination shows a firm, 3-cm swelling on the neck that moves with swallowing; there is no lymphadenopathy. Cardiopulmonary examination shows no abnormalities. Laboratory studies show:

- Hemoglobin 13 g/dL - Leukocyte count 9500/mm3
- Platelet count 230,000/mm3
- Na+ 136 mEq/L K+ 3.5 mEq/L Cl- 104 mEq/L
- TSH 2.3 µU/mL Calcitonin 300 ng/dL (Normal < 5 ng/dL)

- An electrocardiogram shows sinus tachycardia.

Which of the following laboratory abnormalities is most likely to be seen?

a) Increased serum gastrin
b) Increased serum cortisol
c) Increased urinary 5-hydroxyindoleacetic acid
d) Increased serum T3 levels
e) Increased plasma metanephrines

Summary



References

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Thank You